**COMPUTER WAREHOUSE GROUP (CWG) PLC**
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **Metering Department** | METERING & ENERGY ENGINEERING | **17th August 2026 – 21st August 2026** | **Week 11** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**Mechanical Meter Coupling, Component Seating & Real-Time Clock Battery Integration**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 17/08/2026 | PCB & Enclosure Matching | Conducted 100% serial integrity audit matching front cover nameplate serials with electronic PCB barcodes. Inspected internal current transformer terminations. |
| **Tuesday** 18/08/2026 | Internal Battery & LCD Integration | Installed long-life 3.6V Li-SOCl2 lithium backup batteries into PCB retention slots for real-time clock (RTC) retention. Mounted front intermediate housing ensuring clear LCD aperture seating. |
| **Wednesday** 19/08/2026 | GPRS / Cellular Modem Coupling | Mounted 4G/GPRS communication daughterboards onto meter motherboards. Fastened internal whip antenna leads and validated SIM card tray retention mechanism. |
| **Thursday** 20/08/2026 | Torque-Calibrated Screw Fastening | Secured outer transparent casing screws using calibrated electric torque screwdrivers (1.2 Nm) to ensure IP54/IP65 ingress protection against moisture and dust. |
| **Friday** 21/08/2026 | Coupling Ledger Batch Reconciliation | Recorded 240 assembled meter units into the physical Coupling Records Ledger; audited serial batch ranges and routed units to the Post-Test station. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (METER COUPLING & COMPONENT LAYOUT HIERARCHY)**

```
[Outer Transparent Cover] --> [Front Nameplate (Serial # Audit)]
   |--> [Sealing Gasket (IP54 Enclosure Ingress Protection)]
   |--> [Metering PCB Motherboard + LCD Display Aperture]
   |--> [3.6V Li-SOCl2 RTC Backup Battery] + [GPRS/4G IoT Daughterboard]
   |--> [Bi-Stable Magnetic Disconnect Latch Relay (100A Rating)]
   |--> [Base Terminal Housing + Micro-switch Cover Tamper Sensor]
```

**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**
*"Learned the critical importance of serial number reconciliation between physical casing and PCB firmware identity to avoid utility billing discrepancies in the Meter Management System (MMS)."*

**INDUSTRY SUPERVISOR**
Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**
SIWES Coordinator Signature

**COMPUTER WAREHOUSE GROUP (CWG) PLC**
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **Metering Department** | METERING & ENERGY ENGINEERING | **24th August 2026 – 28th August 2026** | **Week 12** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**ESD Grounding Setup, Mains Energization, Keypad Verification & Factory-to-Deployment Mode Switching**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 24/08/2026 | ESD Workstation Grounding Setup | Inspected ESD wrist straps, verified < 1.0 MOhm resistance path to common copper earth bar, and initiated static-safe post-test workbench procedures. |
| **Tuesday** 25/08/2026 | Mains Energization & LCD Boot Diagnostics | Connected coupled smart meters to 230V AC / 50Hz bench power supply; checked power-on self-test (POST) sequences, LCD segment illumination, and relay click. |
| **Wednesday** 26/08/2026 | Elastomer Keypad Verification | Sequentially entered diagnostic test shortcodes (#001# for Total Energy, #002# for Instantaneous Voltage, #003# for Current) to confirm audio-tactile elastomer keypad functionality. |
| **Thursday** 27/08/2026 | MMS Optical Firmware Parameterization | Coupled infrared optical probes to meter front read-heads; used Meter Management Software (MMS) to switch operational mode from Factory Mode to locked Deployment Mode. |
| **Friday** 28/08/2026 | Relay Disconnector Operation & Token Acceptance | Generated 20-digit STS test recharge credit tokens via test generator; loaded tokens via keypad to confirm token acceptance and internal latch relay disconnector trip/close. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (POST-TEST BENCH SETUP & MMS MODE STATE MACHINE)**




**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**
*"Mastered the use of proprietary Meter Management Software (MMS) and optical read-heads. Understood how ESD grounding prevents latent dielectric breakdown in microcontroller gates."*

**INDUSTRY SUPERVISOR**
Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**
SIWES Coordinator Signature

**COMPUTER WAREHOUSE GROUP (CWG) PLC**
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **Metering to IT Systems Handover** | METERING & ENERGY ENGINEERING | **31st August 2026 – 4th September 2026** | **Week 13** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**NEMSA Regulatory Compliance Audit, Tamper-Evident Wire Sealing & Palletized Warehouse Transfer**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 31/08/2026 | Finished Cosmetic & Housing Audit | Inspected 300 tested meters for cosmetic flaws, stripped terminal screws, or missing terminal cover screws. Cleaned surface grease using anti-static fluid. |
| **Tuesday** 01/09/2026 | QC Pass & Statutory NEMSA Sticker Application | Affixed internal QC Pass guarantee stickers over enclosure seam joints; applied official statutory NEMSA (Nigerian Electricity Management Services Agency Act 2015) compliance labels. |
| **Wednesday** 02/09/2026 | Serialized NEMSA Poly-carbonate Wire Sealing | Threaded high-tensile stainless steel sealing wire through meter terminal lugs and casing lugs; crimped serialized NEMSA tamper-evident seals using calibrated sealing pliers. |
| **Thursday** 03/09/2026 | CWG Holographic Sealing & User Manual Insertion | Affixed CWG Plc corporate holographic tamper-evident warranty seals; inserted user operating manuals, STS quick-start guides, and wiring schematics into product cartons. |
| **Friday** 04/09/2026 | Packaging Ledger Audit & Departmental Handover | Completed batch ledger documentation reconciling 300 meter serials with NEMSA seal serials; transferred palletized batch to warehouse store. Prepared for rotation to IT Support. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (NEMSA & CWG TAMPER-EVIDENT MULTI-TIER SEALING SCHEME)**

```
+-------------------------------------------------------------------+
| [Internal QC Seal] -> Validates manufacturing & optical calibration
| [NEMSA Sticker]   -> Statutory compliance (NEMSA Act 2015 grid mandate)
| [Wire Crimp Seal] -> Poly-carbonate serialized security wire through lugs
| [CWG Hologram]    -> Factory warranty authentication seal over casing seam
+-------------------------------------------------------------------+
          |=========> Protective Box Packaging with Foam Cushioning
          |=========> User Manual + STS Token Card Enclosure
          |=========> Palletized Warehouse Transfer for Utility Dispatch
```

**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**
*"Learned the legal and technical necessity of statutory NEMSA verification for grid-connected metering equipment in Nigeria, and how multi-tier physical security seals deter meter tampering."*

**INDUSTRY SUPERVISOR**
Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**
SIWES Coordinator Signature

**COMPUTER WAREHOUSE GROUP (CWG) PLC**
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

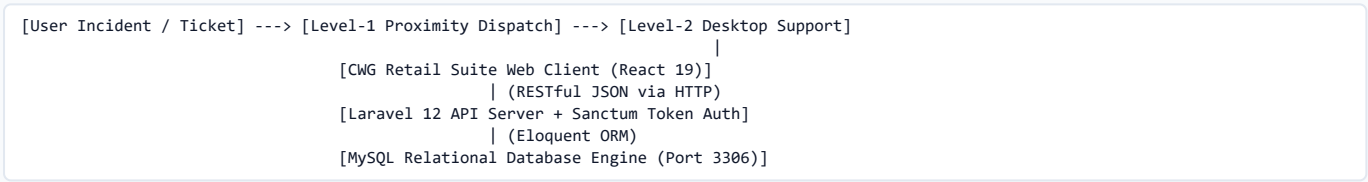
**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **Enterprise IT Infrastructure** | IT SUPPORT & PROXIMITY | **14th September 2026 – 18th September 2026** | **Week 14 (IT Week 1)** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**IT Proximity Helpdesk Orientation, Ticketing SLA Review & POS Retail Suite Architecture Kickoff**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 14/09/2026 | Proximity Unit Orientation & ITSM Workflow | Reported to IT Support & Proximity Unit. Received orientation on enterprise IT service desk operations, SLA priority matrices (Severity 1 to 4), and incident ticketing platforms. |
| **Tuesday** 15/09/2026 | Corporate User Incident Handling | Resolved 8 Level-1 user tickets: diagnosed corporate printer spooler crashes, configured local network shared printers, and assisted staff with Microsoft 365 Outlook profile sync. |
| **Wednesday** 16/09/2026 | CWG Retail Suite Architecture & ERD Review | Participated in software development kickoff meeting for CWG Retail Suite (POS & Inventory). Reviewed system requirements and drafted relational Entity-Relationship Diagram (ERD). |
| **Thursday** 17/09/2026 | Laravel 12 API Scaffold & Sanctum Setup | Installed Laravel 12 backend framework; configured Laravel Sanctum tokenized authentication and structured database connection parameters for MySQL in .env. |
| **Friday** 18/09/2026 | Initial React Frontend Scaffolding & Git Tracking | Scaffolded modern React frontend client using Vite; initialized Git version control repository on GitHub with branch protection rules. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (ITSM ESCALATION & CLIENT-SERVER SYSTEM ARCHITECTURE)**




**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**
*"Gained clear perspective on enterprise helpdesk operations and corporate incident escalation. Successfully initialized the foundational full-stack decoupled architecture for CWG Retail Suite."*

**INDUSTRY SUPERVISOR**
Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**
SIWES Coordinator Signature

**COMPUTER WAREHOUSE GROUP (CWG) PLC**
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

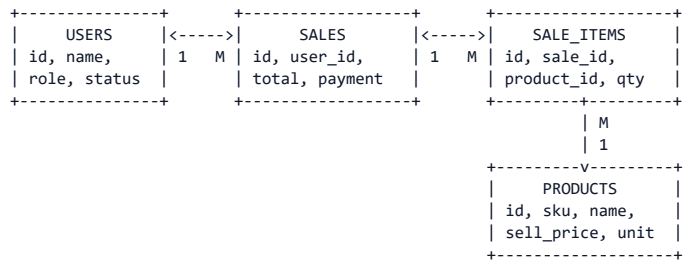
**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **Infrastructure & Systems Development** | SOFTWARE & IT SYSTEMS | **21st September 2026 – 25th September 2026** | **Week 15 (IT Week 2)** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**Enterprise VPN/MFA Support, Database Schema Normalization & Secure RESTful Authentication APIs**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 21/09/2026 | Remote VPN & MFA Configuration | Assisted corporate staff with Zscaler / Cisco AnyConnect secure remote VPN configurations and registered Microsoft Authenticator Multi-Factor Authentication (MFA) tokens. |
| **Tuesday** 22/09/2026 | Database Migrations & Relational Schema (3NF) | Designed and executed Laravel migrations in 3rd Normal Form: `users`, `products`, `sales`, `sale_items`, and `stock_adjustments` tables with foreign key constraints. |
| **Wednesday** 23/09/2026 | Sanctum Token Authentication & Breeze Integration | Implemented user login, password hash verification (Bcrypt), and token issuance endpoints (`POST /api/login`) with role validation attributes. |
| **Thursday** 24/09/2026 | Hardware RAM/SSD Laptop Diagnostic | Diagnosed 2 malfunctioning corporate HP laptops: replaced faulty DDR4 SODIMM RAM modules, upgraded failing HDD to 512GB NVMe SSD, and verified BIOS hardware recognition. |
| **Friday** 25/09/2026 | Product Creation API & Route Testing | Developed `ProductController.php` with validation rules; created API endpoint for product catalogue management and tested JSON responses via automated test suite. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (DATABASE RELATIONAL SCHEMA (3NF) TOPOLOGY)**




**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**
*"Understood the practical intersection between hardware maintenance (SSD/RAM staging) and software data integrity (database normalization and token security)."*

**INDUSTRY SUPERVISOR**
Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**
SIWES Coordinator Signature

## COMPUTER WAREHOUSE GROUP (CWG) PLC

STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **Enterprise Networking & Workstation Maintenance** | IT SUPPORT & NETWORKING | **28th September 2026 – 2nd October 2026** | **Week 16 (IT Week 3)** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**Structured Cabling, Patch Panel Termination & Network Continuity Diagnostics**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 28/09/2026 | Data Center Patch Panel Audit | Audited server rack patch panels on the 2nd floor data room; re-labeled Cat6a drop cables and documented port assignments to Cisco Catalyst 2960 switches. |
| **Tuesday** 29/09/2026 | Cat6 Structured Cabling & RJ-45 Crimping | Terminated 12 new Cat6 solid copper patch cables according to the TIA/EIA-568B standard; verified pinout continuity, wire-map, and near-end crosstalk with digital cable tester. |
| **Wednesday** 30/09/2026 | Cisco Switch VLAN Patching & Trunking | Assisted Senior Network Engineer in patching workstation ports to designated Departmental VLANs (VLAN 10: Admin, VLAN 20: Engineering, VLAN 30: Guest Wi-Fi). |
| **Thursday** 01/10/2026 | Public Holiday: National Independence Day | Statutory National Holiday (Nigeria Independence Day). Conducted scheduled remote server health checks and UPS power status monitoring. |
| **Friday** 02/10/2026 | Wireless Access Point (AP) Deployment | Mounted and configured Aruba Enterprise Wi-Fi Access Points; measured RF signal coverage and resolved channel interference in the executive conference boardroom. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (TIA/EIA-568B COLOR CODE & ENTERPRISE SWITCH PATCHING)**

```
TIA/EIA-568B Pinout Standard:
Pin 1: White/Orange | Pin 2: Orange      | Pin 3: White/Green | Pin 4: Blue
Pin 5: White/Blue   | Pin 6: Green       | Pin 7: White/Brown | Pin 8: Brown
                    |
[Wall Data Outlet RJ-45] ===Cat6 Drop===> [24-Port Patch Panel] ===Patch Cord===> [Cisco Switch (VLAN Tagged)]
```

**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**

*"Mastered precision RJ-45 crimping under TIA/EIA-568B. Gained real-world appreciation for structured cable dressing and cable management in high-density rack enclosures."*

**INDUSTRY SUPERVISOR**

Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**

SIWES Coordinator Signature

## COMPUTER WAREHOUSE GROUP (CWG) PLC

STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **Identity & Access Management** | ACTIVE DIRECTORY ADMINISTRATION | **5th October 2026 – 9th October 2026** | **Week 17 (IT Week 4)** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**Active Directory Domain Services, User Provisioning, OUs & Group Policy Object (GPO) Deployment**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 05/10/2026 | Active Directory User Provisioning | Created new corporate user accounts in Windows Server 2022 Active Directory Users and Computers (ADUC); assigned complex password policies and primary email aliases. |
| **Tuesday** 06/10/2026 | Organizational Units (OUs) & Departmental Groups | Structured directory tree hierarchy: segregated user objects into nested OUs (`CWG-Staff`, `Metering-Dept`, `IT-Operations`, `Finance`); assigned security group memberships. |
| **Wednesday** 07/10/2026 | Group Policy Objects (GPOs) Configuration | Configured and linked Domain GPOs: enforced 15-minute screen lock with password, restricted USB mass storage write access, and mapped departmental network drives (`Z:\Enterprise_Data`). |
| **Thursday** 08/10/2026 | Domain Joining & Kerberos Trust Diagnostics | Joined 6 new corporate Dell Latitude laptops to domain `cwg.local`; verified DNS server lookups, computer account registration, and Kerberos ticket generation. |
| **Friday** 09/10/2026 | Audit of Inactive AD Accounts & Password Resets | Executed PowerShell AD audit scripts to identify user accounts inactive for > 60 days; processed 14 self-service password reset requests from enterprise staff. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (ACTIVE DIRECTORY HIERARCHY & GPO INHERITANCE FLOW)**

```
Domain: cwg.local
  |--> OU: Corporate Head Office
         |--> OU: IT-Operations -------> [GPO: Admin Access & Remote Management]
         |--> OU: Metering-Division ---> [GPO: Optical Testing Hardware Drivers]
         |--> OU: Commercial Staff ----> [GPO: USB Write Block & Inactivity Lock]
                 |--> Users: [staff1, staff2] (Members of 'Domain Users' + 'POS-Operators')
```

**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**
*"Gained in-depth hands-on experience managing Windows Server Active Directory. Understood how GPOs standardize security policies across hundreds of networked endpoints simultaneously."*

**INDUSTRY SUPERVISOR**
Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**
SIWES Coordinator Signature

**COMPUTER WAREHOUSE GROUP (CWG) PLC**
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **Core POS Engine & Inventory Architecture** | SOFTWARE ENGINEERING | **12th October 2026 – 16th October 2026** | **Week 18 (IT Week 5)** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**Product Inventory Normalization, Stock Adjustments Engine & Role-Based Middleware Security**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 12/10/2026 | Product Deactivation & Route Model Binding | Refactored `ProductController` to use Laravel Route Model Binding; added soft-delete `is_active` flag to archive products safely without breaking historical sales data. |
| **Tuesday** 13/10/2026 | Decoupled Inventories Table Migration | Separated stock quantity into dedicated `inventories` table with 1-to-1 product relationship; migrated existing product inventory data seamlessly. |
| **Wednesday** 14/10/2026 | Stock Adjustment Controller & Audit Routing | Implemented `StockAdjustmentController.php` supporting stock audit reasons (`damaged`, `restock`, `inventory_audit`); built transaction logging for stock movements. |
| **Thursday** 15/10/2026 | Multi-Product Automated Feature Tests | Authored Pest/PHPUnit automated test suites (`StockAdjustmentTest.php`) verifying inventory increments, boundary conditions, and negative stock prevention. |
| **Friday** 16/10/2026 | Role Middleware & Cashier Restrictions | Created custom `EnsureUserHasRole` middleware; ensured cashiers are strictly barred from viewing product cost prices, margin data, or store settings. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (INVENTORY AUDIT LOGGING & ROLE GATE STATE MODEL)**

```
Request -> [Role Middleware Check]
             |--> Role == Cashier  ---> Returns: sell_price, sku, name (cost_price HIDDEN)
             |--> Role == Owner/Mgr --> Returns: cost_price, profit_margin, full_audit
                                    |
[Stock Adjustment] ---> [DB Transaction Lock] ---> [Adjust Inventory Qty] ---> [Log Audit Event]
```

**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**
*"Learned professional test-driven development (TDD) using automated PHPUnit assertions. Successfully protected sensitive business financial metrics using server-side role middleware."*

**INDUSTRY SUPERVISOR**
Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**
SIWES Coordinator Signature

**COMPUTER WAREHOUSE GROUP (CWG) PLC**
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

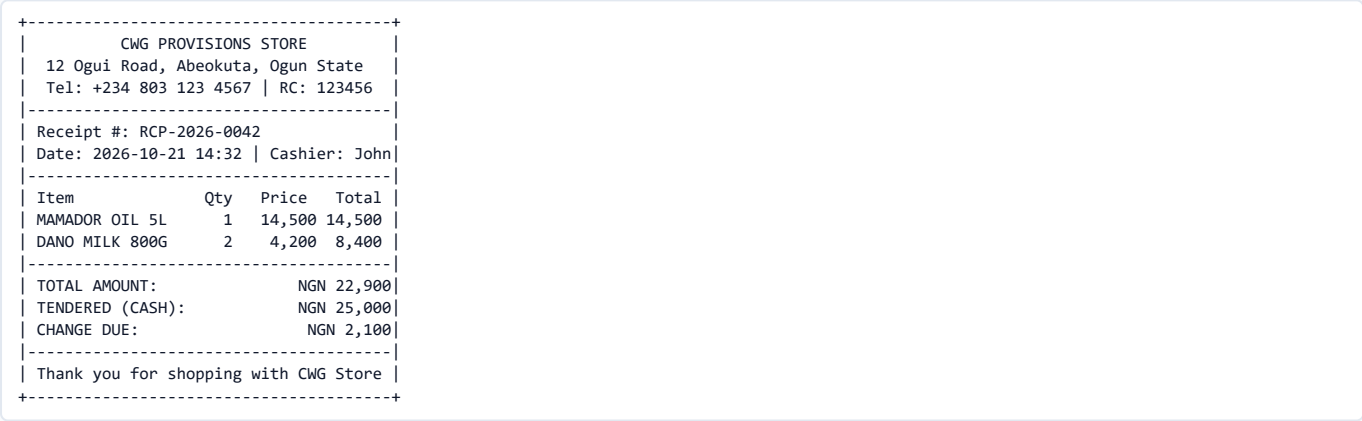
**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **POS Checkout, Thermal Printing & Sales Engine** | SOFTWARE ENGINEERING | **19th October 2026 – 23rd October 2026** | **Week 19 (IT Week 6)** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**POS Checkout Interface, Barcode SKU Search, 80mm Thermal Receipt Generator & Sales History**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 19/10/2026 | POS Interface Grid & Real-Time Cart State | Built React POS component with live category filtering, rapid barcode/SKU search, responsive product grid, and stock-capped cart line item management. |
| **Tuesday** 20/10/2026 | Atomic Cash Checkout & Change Calculator | Engineered checkout settlement modal supporting Cash, Card, and Bank Transfer tender options with automated change calculation and transactional concurrency locking. |
| **Wednesday** 21/10/2026 | Browser-Native 80mm Thermal Receipt Engine | Implemented browser-native 80mm thermal receipt print stylesheet (`@media print`); styled store header, cashier credentials, line items, VAT, and barcode footer. |
| **Thursday** 22/10/2026 | Sales History Table & Date Range Filter Presets | Created sales audit table with filtering by payment method and date presets (Today, 7D, 30D); integrated receipt reprint modal. |
| **Friday** 23/10/2026 | Database Seeding & Restock Intake Station | Seeded 14 authentic Nigerian retail products with images; developed Restock & Adjustment station with reason-tagged adjustments and visual stock badges. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (80MM POS THERMAL RECEIPT LAYOUT STRUCTURE)**




**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**
*"Built an authentic, high-speed retail checkout flow with strict inventory synchronization. Designed pixel-perfect 80mm thermal receipt layouts that print without browser margins."*

**INDUSTRY SUPERVISOR**
Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**
SIWES Coordinator Signature

**COMPUTER WAREHOUSE GROUP (CWG) PLC**
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **Financial Reports & High-Performance Optimization** | SOFTWARE ENGINEERING | **26th October 2026 – 30th October 2026** | **Week 20 (IT Week 7)** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**Financial Reports Engine, SQL B-Tree Indexing, React-Toastify Popups & Activity Audits**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 26/10/2026 | Database Performance & SQL B-Tree Indexing | Eliminated sequential N+1 query loops; added compound B-Tree indexes on `sales(created_at, payment_method)` and projected columns at SQL level for instant response times. |
| **Tuesday** 27/10/2026 | Financial Analysis Layout & 5 Report Subtabs | Constructed comprehensive Reports module: Financial Sales, Inventory Valuation, Stock Level Deficits, Product Velocity, and User Staff Audits with CSV/PDF export. |
| **Wednesday** 28/10/2026 | Adaptive 30-Day Sales Chart & SweetAlert2 Modals | Organized 30-day/90-day daily sales revenue chart with collision-free interval ticks; replaced old browser alerts with modern themed dialogs and React-Toastify notifications. |
| **Thursday** 29/10/2026 | Staff Activity Audit Log & Admin Layer Isolation | Organized admin-only pages under dedicated `pages/admin` directory; added staff event timeline capturing sales transactions, logins, and inventory restock operations. |
| **Friday** 30/10/2026 | IT Technical Report Metering Instructions Update | Incorporated physical manufacturing work instructions from the metering factory (Coupling, Post-Test ESD Safety, and NEMSA Packaging) into the comprehensive IT Report. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (SQL QUERY OPTIMIZATION: BEFORE VS AFTER B-TREE INDEXING)**

```
BEFORE OPTIMIZATION:
  Query: SELECT * FROM sales WHERE created_at BETWEEN ? AND ?
  Execution: Full Table Scan (ALL) -> 5,000+ rows traversed -> High Server Latency (1,800ms)

AFTER B-TREE INDEXING:
  Schema: INDEX sales_created_payment_idx (created_at, payment_method)
  Execution: Range Index Scan (range) -> Directly fetches matched pointer rows -> Sub-10ms response
```

**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**
*"Learned the deep technical mechanics of database query optimization, compound B-tree index execution plans, and architectural modularity in enterprise web development."*

**INDUSTRY SUPERVISOR**
Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**
SIWES Coordinator Signature

**COMPUTER WAREHOUSE GROUP (CWG) PLC**
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

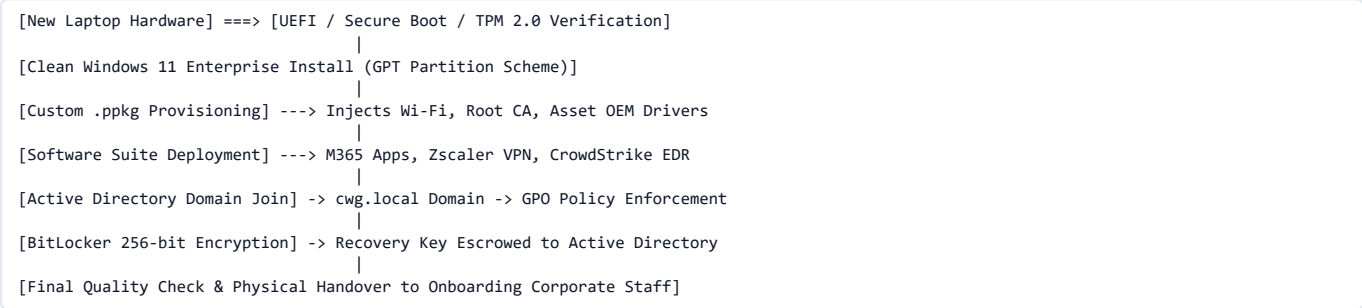
**TRAINING RECORD**
CWG Infrastructure & Engineering

| SECTION / DEPARTMENT | OPERATIONAL UNIT | PERIOD OF TRAINING | LOGBOOK WEEK |
|---|---|---|---|
| **Enterprise IT Operations** | IT SUPPORT & SYSTEMS DEPLOYMENT | **2nd November 2026 – 6th November 2026** | **Week 21 (IT Week 8)** |

**PROJECT / MAIN JOB ASSIGNED DURING THE WEEK**
**Laptop Hardware Staging, Windows OS Provisioning (.ppkg), Essential Software Deployment & Active Directory Domain Integration**

| DAY & DATE | NATURE OF WORK DONE | DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED |
|---|---|---|
| **Monday** 02/11/2026 | New Laptop Unboxing & Hardware POST Verification | Unboxed batch of 5 brand new corporate enterprise laptops (Dell Latitude 5440); verified battery health, verified UEFI BIOS firmware settings, secure boot, and TPM 2.0 status. |
| **Tuesday** 03/11/2026 | Windows 11 Enterprise Clean OS Installation | Configured UEFI bootable flash drives with official Windows 11 Enterprise media; partitioned NVMe SSD storage (GPT scheme) and completed OS base installation. |
| **Wednesday** 04/11/2026 | Automated Staging via Provisioning Packages (.pppkg) | Connected corporate provisioning package (.ppkg) generated with Windows Configuration Designer: automatically injected Wi-Fi profiles, corporate root CA certificates, and OEM drivers. |
| **Thursday** 05/11/2026 | Essential Enterprise Software & Security Suite Installation | Installed corporate software stack: Microsoft 365 Enterprise Apps (Word, Excel, Outlook, Teams), Zscaler Private Access VPN, CrowdStrike Falcon EDR antivirus, and Google Chrome Enterprise. |
| **Friday** 06/11/2026 | Domain Joining, Active Directory Staging & User Profile Handover | Joined newly staged laptops to `cwg.local` Active Directory domain; moved computer objects into `OU=Workstations,OU=CWG-Staff`; tested domain login, BitLocker drive encryption escrow to AD, and handed over to onboarding employee. |

**TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (COMPLETE LAPTOP STAGING & ACTIVE DIRECTORY ONBOARDING WORKFLOW)**




**STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK**
*"Acquired comprehensive practical mastery in end-to-end enterprise laptop staging, automated provisioning with .ppkg packages, essential enterprise security deployment, and Active Directory domain enrollment."*

**INDUSTRY SUPERVISOR**
Signature & Date Stamp

**INSTITUTIONAL SUPERVISOR**
SIWES Coordinator Signature